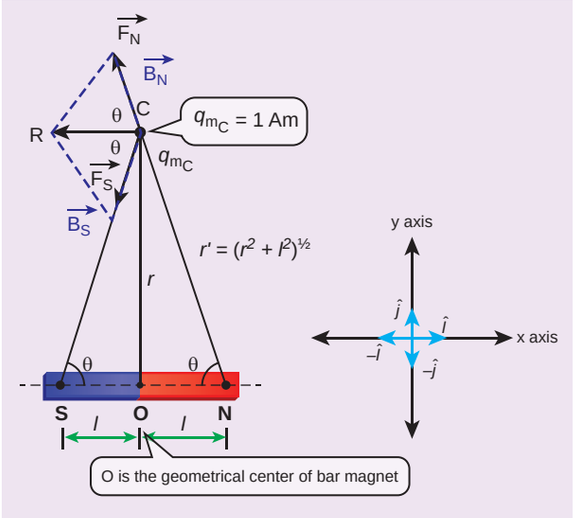


Figure 3.17 Magnetic field at a point along the equatorial line due to a magnetic dipole

The force of repulsion between North Pole of the bar magnet and unit north pole at point C (in free space) is

$$\vec{F}_N = -F_N \cos\theta \hat{i} + F_N \sin\theta \hat{j} \qquad (3.15)$$

where $F_N = \frac{\mu_\circ}{4\pi} \frac{q_m}{r'^2}$

The force of attraction (in free space) between south pole of the bar magnet and unit north pole at point C is (Figure 3.18) is

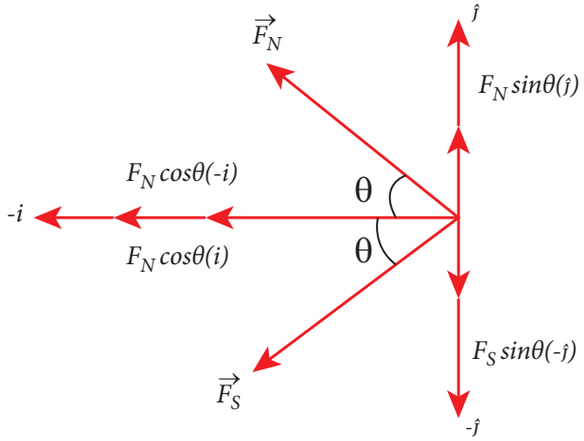


Figure 3.18 Components of force

$$\vec{F}_S = -F_S \cos\theta \hat{i} - F_S \sin\theta \hat{j} \qquad (3.16)$$

where, $F_S = \frac{\mu_\circ}{4\pi} \frac{q_m}{r'^2}$

From equation (3.15) and equation (3.16), the net force at point C is $\vec{F} = \vec{F}_N + F_S$. This net force is equal to the magnetic field at the point C.

$$\vec{B} = -(F_N + F_S)\cos\theta \hat{i}$$

Since, $F_N = F_S$

$$\vec{B} = -\frac{2\mu_\circ}{4\pi} \frac{q_m}{r'^2} \cos\theta \hat{i} = -\frac{2\mu_\circ}{4\pi} \frac{q_m}{(r^2 + l^2)} \cos\theta \hat{i} \qquad (3.17)$$

In a right angle triangle NOC as shown in the Figure 3.17

$$\cos\theta = \frac{adjacent}{hypotenuse} = \frac{l}{r'} = \frac{l}{\left(r^2 + l^2\right)^{\frac{1}{2}}} \qquad (3.18)$$

Substituting equation (3.18) in equation (3.17) we get

$$\vec{B} = -\frac{\mu_\circ}{4\pi} \frac{q_m \times (2l)}{(r^2 + l^2)^{\frac{3}{2}}} \hat{i} \qquad (3.19)$$

Since, magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m \cdot 2l$ and substituting in equation (3.19), the magnetic field at a point C is

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi} \frac{p_m}{(r^2 + l^2)^{\frac{3}{2}}} \hat{i} \qquad (3.20)$$

If the distance between two poles in a bar magnet are small (looks like short magnet) when compared to the distance between geometrical center O of bar magnet and the location of point C i.e., r >>*l*, then,

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